

## Keyword-Based Monoalphabetic Substitution

8. Write a C program for monoalphabetic cipher is that both sender and receiver must commit the permuted cipher sequence to memory. A common technique for avoiding this is to use a keyword from

which the cipher sequence can be generated. For example, using the keyword CIPHER, write out the keyword followed by unused letters in normal order and match this against the plaintext letters:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: C I P H E R A B D F G J K L M N O Q S T U V W X Y Z

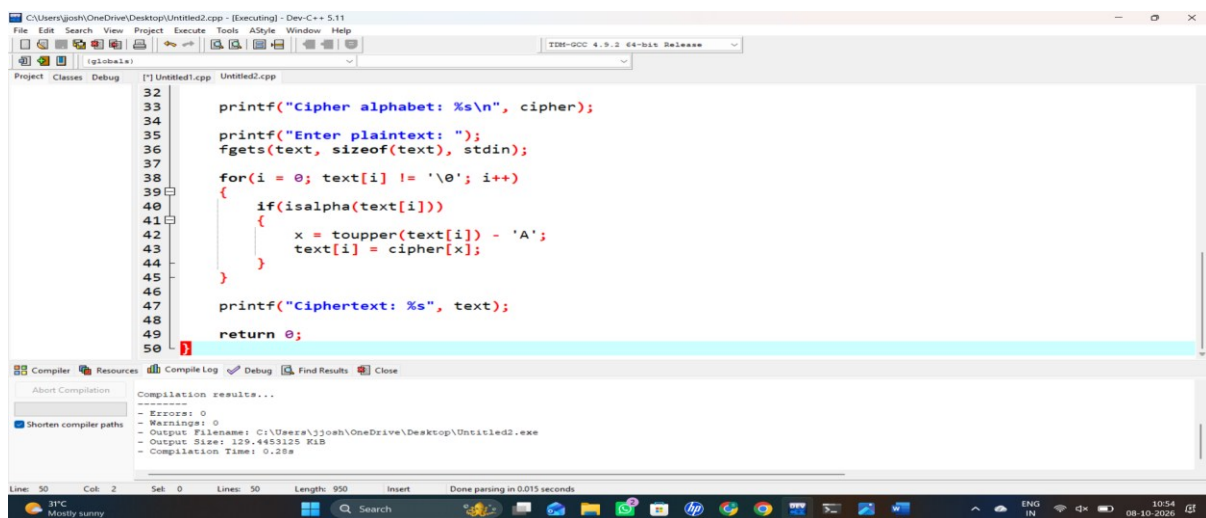
### Aim

To write a C program to implement a keyword-based monoalphabetic substitution cipher.

### Algorithm

1. Read plaintext.
2. Use keyword CIPHER.
3. Remove repeated letters from the keyword.
4. Add remaining unused alphabet letters.
5. Match plaintext alphabet with generated cipher alphabet.
6. Replace each plaintext letter with the corresponding cipher letter.

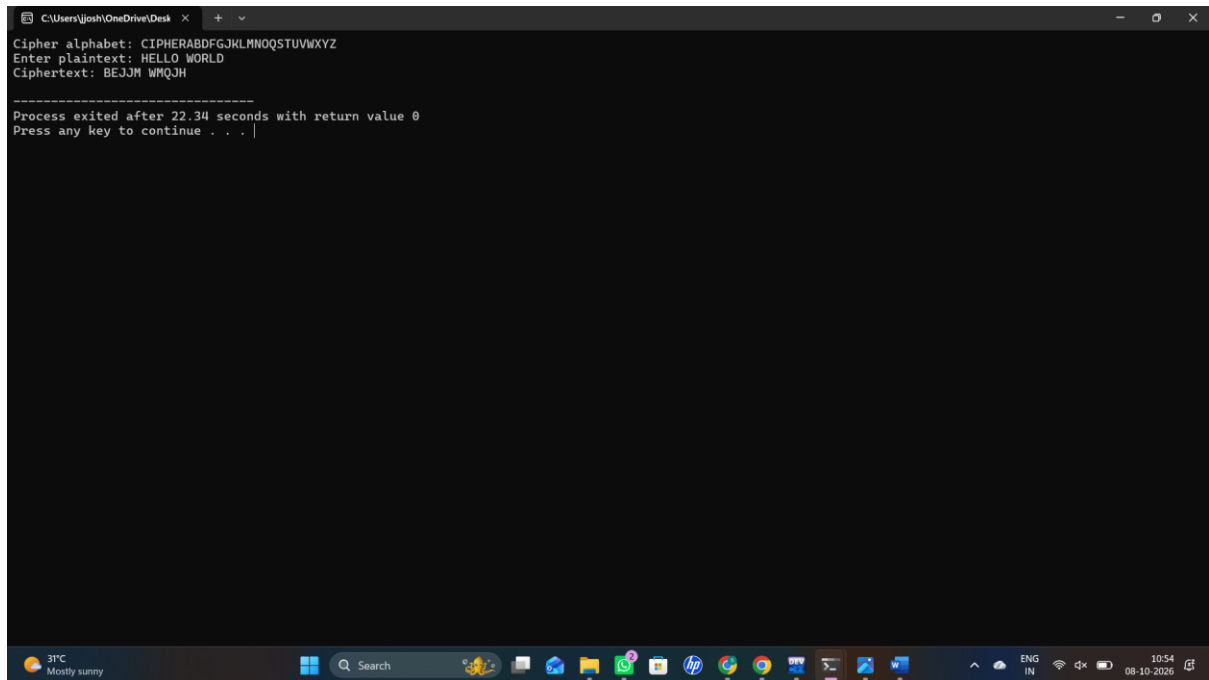
### Program



```
32 printf("Cipher alphabet: %s\n", cipher);
33
34 printf("Enter plaintext: ");
35 fgets(text, sizeof(text), stdin);
36
37 for(i = 0; text[i] != '\0'; i++)
38 {
39     if(isalpha(text[i]))
40     {
41         x = toupper(text[i]) - 'A';
42         text[i] = cipher[x];
43     }
44 }
45
46 printf("Ciphertext: %s", text);
47
48 return 0;
49
50
```

The screenshot shows a C program in Dev-C++ implementing a keyword-based monoalphabetic substitution cipher. The program defines a keyword 'CIPHER' and a cipher alphabet 'CIPHERABDFGJKLMNOPQRSTUVWXYZ'. It reads a plaintext string and replaces each letter with its corresponding cipher letter. The output shows the ciphertext 'CIPHERABDFGJKLMNOPQRSTUVWXYZ'.

# Input/Output



```
C:\Users\josh\OneDrive\Desktop > C:\Users\josh\OneDrive\Desktop\cipher.exe
Cipher alphabet: CIPHERABDFGJKLMNOQRSTUVWXYZ
Enter plaintext: HELLO WORLD
Ciphertext: BEJJM WMQJH

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Process exited after 22.34 seconds with return value 0
Press any key to continue . . .
```

## Result

Thus, the plaintext was successfully encrypted using a keyword-based monoalphabetic substitution cipher.